



Advanced Traffic (20–551)
Class Project Part 6

Part 6: SUMO Simulation with the Same Configuration

Goal:

1. Explore libsumo.
2. Reproduce the Python-based traffic simulation in the SUMO traffic simulator.
3. Use Python to generate all inputs and configuration files.

Tasks

- Learn the basics of SUMO: network definition, route files, and configuration files.
- Convert the previously defined network (nodes, edges, intersections) into SUMO input files.
- Generate vehicle demand using the same logic as in the Python simulation.
- Run the simulation in SUMO and visualize the traffic flow.
- Compare the SUMO results with the custom Python simulation.

Output

1. SUMO input and configuration files (`.net.xml`, `.rou.xml`, `.sumocfg`).
2. Comparison report.
3. Present the code and run the simulation.

Time

Weeks 5–6

Grade Weight

25% (`.net.xml`: 5%, `.rou.xml`: 5%, runs without errors: 5%, correct outputs: 10%)